

Resolving A Vector

SYNOPSIS

Head-to-tail addition joins two vectors into one. *Resolving* a vector is that process in reverse: instead of combining two vectors into a resultant, we split *one* vector into two perpendicular parts — a *horizontal component* and a *vertical component* — that, added head-to-tail, reconstruct the original vector exactly. A vector that already lies flat along an axis needs no resolving at all: it's entirely one component and zero the other. Resolving is the tool that lets us add vectors algebraically instead of just diagrammatically — add the horizontal parts together, add the vertical parts together, and reassemble.

PERFORMANCE EXPECTATIONS By the end of this note, you will be able to...

- PE1 Recognize resolving as the reverse of head-to-tail addition** — splitting one vector into two perpendicular parts instead of combining two into one.
- PE2 Resolve a vector into its horizontal (x) and vertical (y) components** by drawing a right triangle: the component legs run along the axes, and the original vector is the hypotenuse.
- PE3 Recognize when a vector needs no resolving** — a vector already lying along the x-axis or y-axis is entirely one component; its other component is zero.
- PE4 Resolve two or more vectors, then add their components separately** — sum all the x-components together, and all the y-components together.
- PE5 State the component-addition rule** — $C_x = A_x + B_x$ and $C_y = A_y + B_y$ — and explain why adding components independently gives the same resultant as head-to-tail addition.



You already know this

Splitting one diagonal thing into two straight parts isn't new — you've done it before:



TV screen size

A "55-inch" TV is measured along the diagonal, but the screen really has a width and a height. The diagonal is one number standing in for two perpendicular ones.



Cutting across a field

Walk diagonally across a rectangular field instead of around the two edges — you still end up some distance east and some distance north, just covered both at once instead of one leg at a time.



Pushing a joystick diagonally

Push a game controller stick diagonally and the game doesn't see "diagonal" — it resolves your push into an x-push and a y-push and moves your character by both, separately.

In this note, we work through

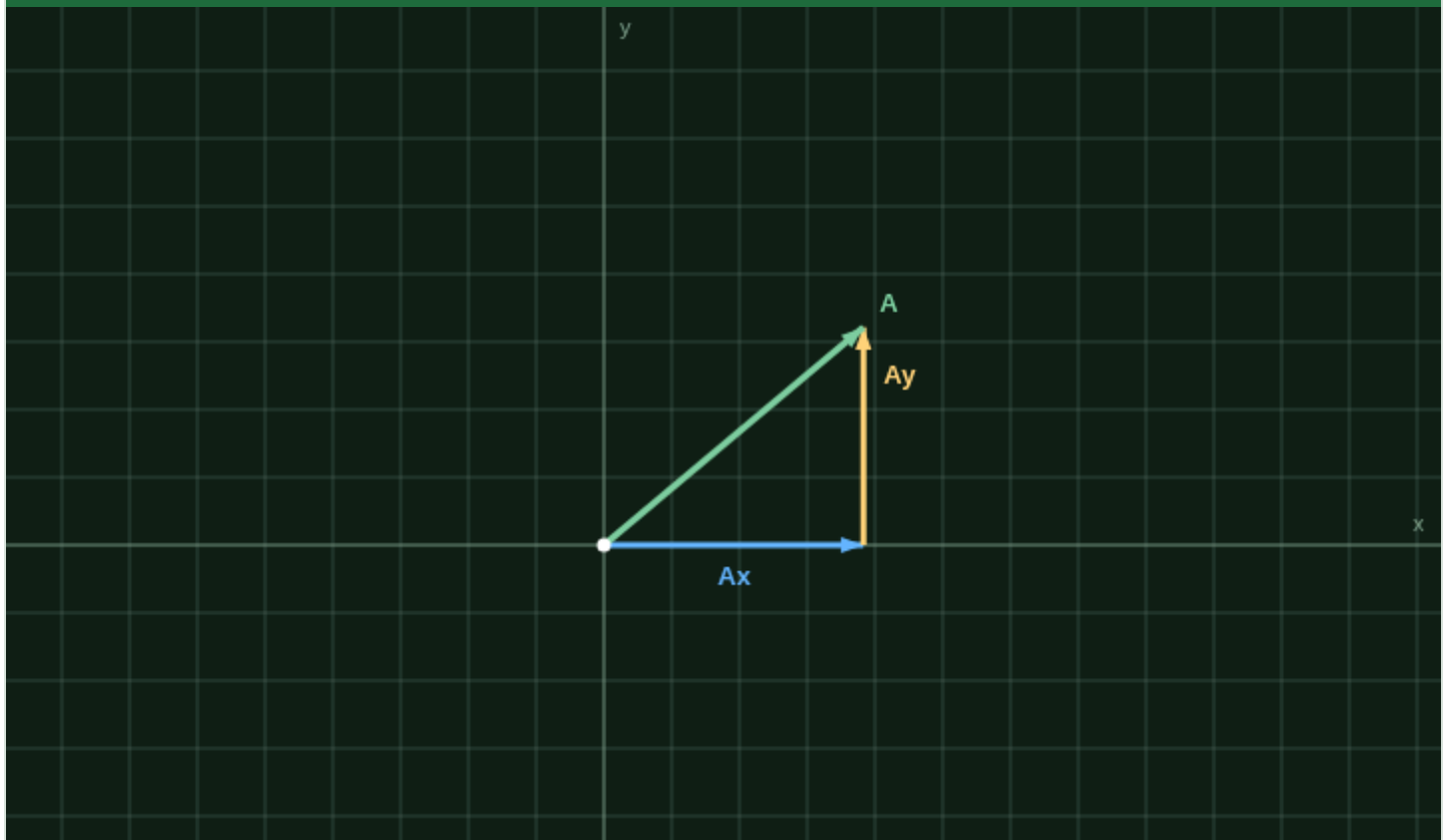
- I. What does it mean to "resolve" a vector, and how is that head-to-tail addition in reverse?
- II. How do we find a vector's horizontal and vertical components?
- II. What happens when we add two vectors' components separately — do we get the same resultant as adding the vectors head-to-tail?
- IV. Why does a vector already lying along an axis need no resolving?

Play with it: resolve a vector into components

Start in **One vector** mode. Set vector **A**'s length and angle. **A_x** is how far it reaches horizontally; **A_y** is how far it reaches vertically. Together they form a right triangle with **A** itself as the hypotenuse. Try dragging the angle to exactly 0° or 90° — watch one component disappear.

Then switch to **$\mathbf{A} + \mathbf{B} = \mathbf{C}$** mode. Two vectors are joined head-to-tail, exactly as in Note 1, and the resultant **C** is drawn dashed from A's tail to B's head. Show the components and compare the chips: **A_x** + **B_x** lands on **C_x**, and **A_y** + **B_y** lands on **C_y** — for *any* two vectors you can set. Change the lengths and angles and try to break it.

► INTERACTIVE · VECTOR COMPONENT RESOLVER



Vector A — length

5.0 m

Vector A — angle

40°

Vector B — length

3.0 m

Vector B — angle

120°

$A_x \approx 3.8 \text{ m}$

$A_y \approx 3.2 \text{ m}$

This lab is interactive. The sliders and buttons only work on screen; the figure above is one state of it, with the settings shown beneath. Open the note online to run it yourself: https://byungmooncho.github.io/PHY151/Module_02/M2_N2_Resolving_A_Vector.html

Predict first: a vector 5 m long points straight up, along the +y axis (90°). What is its horizontal (x) component?

5 m

0 m

2.5 m

It depends on the y-component

Watch: working through the note

WALKTHROUGH VIDEO • ~10 MIN

Watch the walkthrough in the online note.

Component signs at a glance

Before you compute anything, read the signs straight off the arrow's direction. Rightward and upward are positive, exactly as in Module 1:

THE VECTOR POINTS...	X-COMPONENT	Y-COMPONENT
Up and to the right ↗	+	+
Up and to the left ↖	−	+
Down and to the right ↘	+	−
Down and to the left ↙	−	−
Exactly along an axis → or ↑	one component is the whole vector; the other is zero	

Looking ahead: in this note we resolve vectors by *drawing* — building the right triangle and reading the legs off the grid. In Note 2a, trigonometry (SOH-CAH-TOA) gives the exact component lengths for any angle, no grid required. That's the tool the projectile notes (N3-N5) run on.

Three rules to remember



$$\mathbf{A} \rightarrow \mathbf{A}_x + \mathbf{A}_y$$

Resolving splits one vector into a horizontal part and a vertical part that rebuild it head-to-tail.



$$\mathbf{C}_x = \mathbf{A}_x + \mathbf{B}_x$$

Components of the resultant are the sums of the components — and the same along y: $C_y = A_y + B_y$.



Vectors → numbers

The payoff: with components, vector addition becomes ordinary addition, done separately along x and along y.

Exit ticket

Four quick questions — check yourself before you close this note.

Q1. Resolving a vector into components means...

- ☐ Rotating it 90°.
- ☐ Splitting it into a horizontal part and a vertical part that add back to the original vector.
- ☐ Doubling its length.
- ☐ Combining it with another vector.

Q2. A vector lies exactly along the +x axis. Its vertical (y) component is...

- ☐ Equal to its magnitude.
- ☐ Zero.
- ☐ Impossible to determine.
- ☐ Always negative.

Q3. Vector A has $A_x = 6$ m and vector B has $B_x = -2$ m. If $C = A + B$, what is C_x ?

m

Q4. Adding vectors by components — $C_x = A_x + B_x$ and $C_y = A_y + B_y$ — works because...

- ☐ The horizontal and vertical directions are independent of each other, so components along each axis add like ordinary numbers.
- ☐ It's a shortcut that only works for right angles between A and B.
- ☐ It only works when A and B point the same way.
- ☐ Vectors don't actually add — only their components do.

Your notes

Your notes

Guided notes — blank (PDF):

https://byungmooncho.github.io/PHY151/handouts/M2_N2_Resolving_A_Vector_Handout.pdf

Completed notes (PDF): released when Module 2 closes.

